Full support and positive density in the zero-temperature SK model

A reconstruction of OpenAI's quantitative argument · 26 October 2026

Abstract. The Stieltjes measure of every integrable minimiser of the zero-temperature Parisi functional for the pure, zero-field Sherrington–Kirkpatrick model has full support on $[0,1)$ and a smooth density that is strictly positive on $(0,1)$. We reconstruct OpenAI's quantitative argument, organised around two uses of one spatial-moment inequality. On a positive constant interval, it makes the derivative of the gradient second moment strictly convex, contradicting the endpoint conditions imposed by minimality. The same curvature and the bounded gradient also exclude terminal gaps. Full support then gives moment identities and smoothness; a local constant replacement transfers the inequality to the minimiser and forces its density to be positive. H.-B. Chen previously proved full support and smooth nonnegative density. We isolate the fixed-time spatial moment and the positive margin that survives local replacement, while importing its shape comparisons and full polynomial identity. We also derive the scalar value identity and state the classical optimisation consequence and the additional steps leading to asymptotically optimal quantum approximate optimization algorithm (QAOA) circuits.

1. The result and the mechanism

For independent standard Gaussian couplings J_{ij}, the pure, zero-field Sherrington–Kirkpatrick (SK) Hamiltonian is

$$H_n(\sigma) = n^{-1/2} \sum_{i<j} J_{ij}\sigma_i\sigma_j, \qquad \sigma \in \{-1,1\}^n.$$

The limiting ground-state energy per spin, with the maximisation convention, is

$$P_* := \lim_{n\to\infty} n^{-1}\mathbb{E}\max_{\sigma} H_n(\sigma).$$

The Parisi formula expresses P_* as a minimum over nondecreasing functions. Our subject is the structure of a minimiser: its increments charge every interval of overlap scales below one, and in fact have strictly positive density at each interior scale. The proof uses one quantitative bound twice, first to exclude constant intervals and then, after an approximation, to prove pointwise positivity.

For two configurations, their overlap is $R(\sigma,\tau) = n^{-1}\sum_i \sigma_i\tau_i$. It determines the energy covariance: $\mathbb{E}[H_n(\sigma)H_n(\tau)] = nR^2/2 - 1/2$. Thus the Gaussian law of the energies is encoded by the overlaps. We use the covariance convention $\xi(t) = t^2/2$, so $\xi'' = 1$; adding one independent centred Gaussian of variance $1/2$ to every configuration's energy gives covariance $n\xi(R)$ and leaves the expected maximum unchanged. This is the normalisation of the ground-state Parisi formula of Auffinger and Chen (2017), Theorem 1. There are no higher-order interactions and no external field.

Let $\mathcal{U}$ consist of the nonnegative, nondecreasing, right-continuous functions $\gamma : [0,1) \to [0,\infty)$ satisfying $\int_0^1 \gamma(t)\,dt < \infty$. Define the locally finite Stieltjes measure by

$$\mu_\gamma([0,t]) = \gamma(t), \qquad 0 \le t < 1.$$

Thus $\mu_\gamma((s,t]) = \gamma(t) - \gamma(s)$: the measure records the increments of γ. A point mass (atom) of size $\gamma(0)$ is allowed at zero, and the total mass need not be finite or equal to one. All support statements use the relative topology of $[0,1)$. A support gap is a relatively open interval of positive length with zero mass; the cumulative coefficient is constant on such an interval.

The Parisi functional $\mathcal{P}(\gamma)$ is defined using an auxiliary value function $\Phi_\gamma(t,x)$, where $x \in \mathbb{R}$ is a spatial variable and t is the overlap parameter, treated as time in the equation. Write $\Phi = \Phi_\gamma$ and use subscripts for partial derivatives. The backward equation, terminal condition and functional are

$$\Phi_t = -\tfrac{1}{2}(\Phi_{xx} + \gamma(t)\Phi_x^2), \qquad \Phi(1,x) = |x|, \qquad \mathcal{P}(\gamma) = \Phi_\gamma(0,0) - \tfrac{1}{2}\int_0^1 t\gamma(t)\,dt.$$

The terminal value $|x|$ is the zero-temperature value of one spin in a field x. For an integrable coefficient, the PDE solution is interpreted by approximation from bounded monotone coefficients. The exact control definition and stability theorem are OpenAI (2026a), §1 and Lemmas 2.1–2.2. To turn minimality of $\mathcal{P}$ into a condition on γ, we use a random process whose gradient moment measures the first-order cost of changing γ. Write $u = \Phi_x$ for the spatial gradient and $a = \Phi_{xx}$ for its spatial derivative. Let B be standard Brownian motion and define the diffusion X and the moment q by

$$dX_t = \gamma(t)u(t,X_t)\,dt + dB_t, \qquad X_0 = 0, \qquad q(t) = \mathbb{E}[u(t,X_t)^2].$$

The discrepancy $q(t) - t$ enters the first variation of $\mathcal{P}$; minimality will force $q(t) = t$ on the support of μ_γ. The SDE has a unique strong solution. All stochastic calculations below take place on compact time intervals ending before one, unless an endpoint limit is justified explicitly.

Auffinger, Chen and Zeng (2020), Theorem 1 proved infinite support. This need not be full support: the measure $\sum_{k\geq 1} 2^{-k}\delta_{1-2^{-k}}$ has infinitely many atoms and many gaps. H.-B. Chen (2026), Theorem 1.3 subsequently proved full support and smooth nonnegative density. We follow OpenAI's quantitative proof of the support conclusion and its further strict-positivity result. The feature used twice is a lower bound valid throughout every positive constant interval of an admissible coefficient, without a minimising assumption.

Theorem 1 (full support; quantitative proof following OpenAI (2026a), Theorem 1.1). If $\gamma \in \mathcal{U}$ minimises $\mathcal{P}$, then

$$\operatorname{supp}\mu_\gamma = [0,1), \qquad \gamma(t_2) > \gamma(t_1) \quad (0 \leq t_1 < t_2 < 1).$$

For the solution of the displayed SDE and every $0 \leq t < 1$,

$$q(t) = t, \qquad \mathbb{E}[a(t,X_t)^2] = 1.$$

Corollary 4 in §4 gives the smooth density and its strict positivity at every interior point.

Existence of a minimiser is a separate input from Auffinger and Chen (2017), Theorem 1 and the proof of Lemma 3. Integrability allows γ to grow near one. Zero field gives $u(0,0) = 0$, and $\xi'' = 1$ is used in the curvature calculation; the theorem here does not extend these hypotheses to nonzero field or a general mixed-spin covariance.

Our purpose is to isolate the spatial moment common to the support and density arguments, explain why it needs an integrated estimate, and identify what survives the local-replacement limit. The conclusions and the quantitative method are those of the cited sources. The proof compares two demands on a gap with two support endpoints. Minimality forces q to agree with time there, with endpoint slopes at most one. On a positive constant interval a quantitative estimate makes q' strictly convex, so its integral is smaller than the interval's length. This contradicts the required change in q. After excluding all gaps, the identity $q' = 1$ supplies the equation from which the same

estimate will force $\gamma' > 0$. This second use needs the estimate for a local constant replacement that need not minimise, without requiring $q'' = 0$ at the comparison time. Its explicit positive margin must also survive the replacement's limit.

Section 5 derives the scalar value formula and identifies the finite Gaussian observable used by the separate QAOA theorem. These optimisation consequences use full support; strict density positivity is a further analytic conclusion. The analytic argument assumes Itô calculus and martingale convergence. The circuit construction in §5.3 is a separate imported theorem; we identify its finite input and the energy it approximates. Omitted proofs are linked to *From Parisi full support to deterministic QAOA circuits: A long reasoning record*, the companion edition of 26 October 2026; original-source citations remain beside the imported inputs.

2. Minimality gives contact; curvature excludes gaps

For an admissible coefficient, Φ is even, convex and one-Lipschitz in space. Its positive-order spatial derivatives are bounded and continuous on each strip $[0,T] \times \mathbb{R}$, $T < 1$; the PDE holds almost everywhere in time. The compact-strip approximation theorem justifies the following Itô calculations even when γ has jumps (see OpenAI (2026a), Lemmas 2.1–2.2). The control method has antecedents in Jagannath and Tobasco (2016) at positive temperature and Auffinger and Chen (2017), Theorem 2 and Corollary 2 at zero temperature; see also Chen, Handschy and Lerman (2018), Proposition 2 and Theorem 5.

Fix a minimiser. Itô's formula cancels the drift of $u(t, X_t)$:

$$du(t, X_t) = a(t, X_t)\, dB_t, \qquad q(0) = 0, \qquad q'(t) = \mathbb{E}[a(t, X_t)^2].$$

In particular q' is continuous across coefficient jumps. Minimality means that every allowed one-sided change has nonnegative first-order cost. The one-sided variation formula gives the rate of change of the functional when γ is replaced by $\gamma + \varepsilon\eta$ as $\varepsilon \downarrow 0$:

$$D_+\mathcal{P}(\gamma)[\eta] = \tfrac{1}{2}\int_0^1 \eta(t)(q(t) - t)\, dt, \tag{1}$$

for every integrable direction η for which $\gamma + \varepsilon\eta \in \mathcal{U}$ for all sufficiently small $\varepsilon \geq 0$ (see OpenAI (2026a), Lemma 6.1). Earlier zero-temperature variation and support identities appear in Chen, Handschy and Lerman (2018), Propositions 3–4.

Adding a unit atom at s raises the cumulative function at every $t \geq s$, so its direction is $\eta = \mathbf{1}_{[s,1)}$. The first-order cost is therefore a tail integral:

$$G(s) := \int_s^1 (q(t) - t)\, dt \geq 0.$$

Here $G(s)/2$ is the first-order cost of the added atom. Conversely, let $\nu = \mu_\gamma|_K$ for a compact $K \subset [0,1)$. For $0 \leq \varepsilon \leq 1$, the measure $\mu_\gamma - \varepsilon\nu = (1 - \varepsilon\mathbf{1}_K)\mu_\gamma$ is nonnegative, and its cumulative function is $\gamma - \varepsilon\nu([0,\cdot]) \in \mathcal{U}$. Minimality gives

$$0 \leq D_+\mathcal{P}(\gamma)[-\nu([0,\cdot])] = -\tfrac{1}{2}\int_K G(s)\, d\mu_\gamma(s).$$

The change of integration order is justified by $|q - t| \leq 1$ and $\mu_\gamma(K) < \infty$. Since $G \geq 0$, it vanishes almost everywhere for μ_γ, and continuity makes it vanish on $S = \operatorname{supp}\mu_\gamma$. The variation removes a fraction of all mass in K, not only atoms; no density or probability constraint is assumed.

At a support point $s > 0$, G has a minimum. Hence

$$G'(s) = s - q(s) = 0, \qquad G''(s) = 1 - q'(s) \geq 0, \quad \text{so} \quad q(s) = s, \quad q'(s) \leq 1. \tag{2}$$

If $0 \in S$, contact gives $G(0) = 0$; independently, zero field gives $q(0) = 0$. Hence $G(h) = \frac{1}{2}(1 - q'(0))h^2 + o(h^2)$; nonnegativity gives the same slope bound at zero. These are the contact conditions of OpenAI (2026a), Lemma 6.2.

Lemma 2 (constant-interval curvature; OpenAI (2026a), Proposition 5.1). Let $\gamma \in \mathcal{U}$, with its associated diffusion and moment q. If $\gamma = c$ on an open interval $I \subset (0, 1)$, called a plateau, then for $t \in I$

$$q'''(t) \geq \frac{c^2}{1000}\mathbb{E}[a(t, X_t)^4] > 0 \quad \text{if } c > 0. \tag{3}$$

If $c = 0$, then $q''(t) = \mathbb{E}[\Phi_{xxx}(t, X_t)^2] > 0$.

Minimality is not a hypothesis of this lemma. Section 3 derives its moment polynomial and explains the integrated estimate imported from the source.

Proof of Theorem 1, assuming Lemma 2. A complementary interval with two support endpoints admits two contact conditions; an initial interval without support at zero has coefficient zero. For terminal gaps, the bound $q \leq 1$ replaces the missing right contact condition.

Suppose (ℓ, r) is a support gap with endpoints in S and $r < 1$. The coefficient is a positive constant there: support at ℓ and absence of mass immediately to its right force $\mu_\gamma([0, \ell]) > 0$. Strict convexity of q' and (2) give

$$q'(t) < \frac{r - t}{r - \ell}q'(\ell) + \frac{t - \ell}{r - \ell}q'(r) \leq 1 \qquad (\ell < t < r).$$

Integrating contradicts $q(r) - q(\ell) = r - \ell$. The geometric obstruction is an area deficit under q'. An initial gap $[0, r)$ with $0 \notin S$ and $r \in S$ has no mass, hence $\gamma = 0$ there. Its right endpoint gives $q(r) = r$ and $q'(r) \leq 1$, while zero field gives $q(0) = 0$. The zero-plateau identity gives $q'' > 0$, hence $q'(t) < q'(r) \leq 1$. The same integral contradiction follows without assuming $G(0) = 0$. A gap $(0, r)$ with $0 \in S$ is already covered by the two-contact case.

A terminal gap $(r, 1)$ with nonempty support has $r \in S$ and positive plateau height. Suppose $q(t) \geq t$ for some $r < t < 1$. Strict convexity of q' and $q'(r) \leq 1$ give

$$t - r \leq \int_r^t q'(s)\, ds < \frac{t - r}{2}(q'(r) + q'(t)),$$

so $q'(t) > 1$. Convexity then gives $q'(s) > q'(t) > 1$ for $s > t$. Consequently $q(s) > t + (s - t)q'(t) > 1$ for $s < 1$ sufficiently close to one, contrary to $q \leq 1$. Thus $q(t) < t$ on $(r, 1)$, and $G(r) < 0$ contradicts minimality. This uses no contact at one.

If S is empty, then $\gamma = 0$. Lemma 2 makes q strictly convex, and $q(0) = 0$, $q \leq 1$ give, for $0 < t < b < 1$, $q(t) < (t/b)q(b) \leq t/b$. Letting $b \uparrow 1$ gives $q(t) \leq t$. Applying strict convexity again with $t < b < 1$ now gives $q(t) < (t/b)q(b) \leq t$, so $G(0) < 0$.

The terminal exclusion was already recorded by Auffinger, Chen and Zeng (2020), Remark 7. The argument here uses only the curvature needed for the other gaps; long §5, "A shorter qualitative exclusion" gives the full endpoint discussion. The stronger boundary estimate $1 - q(t) \geq C\sqrt{1 - t}$ for arbitrary coefficients constant near one is proved in OpenAI (2026a), Lemma 6.3 and long §5, terminal boundary estimate; Chen (2026), Proposition 4.5, equations (4.27)–(4.38) gives the

square-root asymptotic under the terminal-gap hypothesis. The zero-coefficient heat argument and late-step variation have antecedents in Chen, Handschy and Lerman (2018), Theorem 6.

These cases exhaust the complement of the relatively closed set S. Consequently $S = [0, 1)$, contact gives $q(t) = t$, and differentiation gives $\mathbb{E}a^2 = 1$, including the right-hand value at zero. Moreover, $\gamma(t_2) - \gamma(t_1) = \mu_\gamma((t_1, t_2]) > 0$. This proves Theorem 1 using the stated analytic inputs.

3. The integrated estimate

The quantity used in both arguments is a spatial moment. Write $b = \Phi_{xxx}$ and $d = \Phi_{xxxx}$, and, for $0 < t < 1$, set

$$\mathcal{C}_\gamma(t) = \mathbb{E}\big[d^2 - 12\gamma(t)ab^2 + 6\gamma(t)^2a^4\big],$$

with the spatial factors evaluated at (t, X_t). Its definition requires spatial regularity but no time derivative of γ. The estimate we need can therefore be stated at one time.

Proposition 3 (fixed-time spatial-moment estimate). Let $\gamma \in \mathcal{U}$, let $0 < t < 1$ be a continuity point of γ, and suppose that $c = \gamma(t) > 0$. For its associated diffusion,

$$\mathcal{C}_\gamma(t) \geq \frac{c^2}{1000}\mathbb{E}[a(t, X_t)^4].$$

Neither minimality nor a constant neighbourhood of t is required. This is the consequence of OpenAI (2026a), Proposition 5.1 and Lemmas 2.1–2.2 stated in the companion's Proposition 4.1, §4.5. We first explain the positive-plateau estimate, then obtain this fixed-time form by inserting a short plateau.

For a smooth coefficient the identity connecting the two applications is

$$q''' + 2\gamma'\mathbb{E}a^3 = \mathcal{C}_\gamma.$$

On a plateau, $\gamma' = 0$; at a smooth minimiser, full support gives $q = t$ and hence $q''' = 0$. Section 4 proves the required smoothness before using the second identity.

On a plateau with $\gamma = c > 0$, put $r = c\Phi_x = cu$, $v = r_x = ca$, $w = r_{xx}$ and $z = r_{xxx}$. This rescaling makes the derivative identities polynomial. The PDE becomes $r_t = -\frac{1}{2}r_{xx} - rr_x$, and the diffusion has spatial generator $\frac{1}{2}\partial_{xx} + r\partial_x$. Differentiating in space and applying Itô's formula gives the drift and diffusion pairs $(0, v)$, $(-v^2, w)$ and $(-3vw, z)$ for r, v, w, respectively. All expectations in this section evaluate spatial functions at (t, X_t). Starting with $c^2q' = \mathbb{E}v^2$, Itô's formula gives

$$\begin{aligned} c^2q'' &= \mathbb{E}[w^2 - 2v^3], \\ c^2q''' &= \mathbb{E}[z^2 - 12vw^2 + 6v^4] = c^2\mathcal{C}_\gamma(t). \end{aligned} \tag{4}$$

For example, the drifts of w^2 and v^3 are $z^2 - 6vw^2$ and $3vw^2 - 3v^4$, respectively. The negative mixed term is the obstacle. Convexity, which gives $v \geq 0$, does not control it. Indeed, for the smooth auxiliary terminal value $\log(2\cosh x)$ and coefficient $\gamma \equiv 1$, the solution is $\Phi(t, x) = \log(2\cosh x) + (1 - t)/2$. Thus $r = \tanh x$; at $r^2 = 1/2$, the polynomial in (4) equals $-19/8$. This is a finite-step smoothing example, not a minimiser of the absolute-value-terminal problem; its calculation is given in long §4.1. The same section gives a negative pointwise value with the actual $|x|$ terminal datum for a small positive constant coefficient (long §4.1, pointwise obstruction).

The proof subtracts a weighted divergence whose expectation under the law of X_t is zero. The resulting expression has a positive fourth-moment remainder. The backward value function and

the forward density must satisfy joint shape restrictions for this decomposition to have the required signs. Those comparisons and the exact polynomial identity are the substantial inputs from OpenAI (2026a), §§3–4. We explain their roles first for finite-step coefficients, then pass the integrated inequality to an integrable coefficient by approximation.

Begin with a positive nondecreasing finite-step coefficient γ_{step}: a function constant on each interval of a finite partition. Replace the terminal value $|x|$ by the smooth approximation $M^{-1}\log(2\cosh Mx)$, with $M \geq \|\gamma_{\mathrm{step}}\|_\infty$. On a step of height c, set $h = e^{c\Phi}$. Then $h_t = -h_{xx}/2$ and $r = h_x/h$. The same exponential removes the drift from the forward density equation. Indeed, writing $f = hg$ gives

$$\begin{aligned} f_t &= \tfrac{1}{2}(hg)_{xx} - (h_x g)_x \\ &= -\tfrac{1}{2}h_{xx}g + \tfrac{1}{2}hg_{xx} = h_t g + \tfrac{1}{2}hg_{xx}. \end{aligned}$$

Comparison with $f_t = h_t g + hg_t$ yields $g_t = g_{xx}/2$. Thus one backward heat factor and one forward heat factor determine the law against which the moment polynomial is integrated. For $s = -\partial_x \log g$, the source proves joint shape restrictions: on $x > 0$, $r_x > 0$ and $s_x \geq 1/t$, both r and s are concave, and $H := w/(2r) + v \leq 0$ is nondecreasing. They also give

$$j := s_x - (s/r)v = r\,\partial_x(s/r) \geq 0.$$

The last inequality couples the backward gradient to the forward density; separate monotonicity and concavity do not supply it. On the first step, $s = x/t$, so concavity of r gives $j \geq 0$. A parabolic comparison propagates the sign within each step. At a forward jump with $k = c_+/c_- \geq 1$, continuity of Φ and f gives

$$r_+ = kr_-, \qquad s_+ = s_- + (k-1)r_-, \qquad j_+ = (s_x)_+ - \frac{s_+}{r_+}(r_x)_+ = j_-.$$

The joint sign therefore survives coefficient jumps as well. The quotients w/r and s/r extend smoothly through zero because r, s, w are odd and $r_x(0) > 0$. The comparison's boundary controls and these extensions are analytic inputs from OpenAI (2026a), Lemmas 3.2–3.4, reconstructed in long §4.3.

The factorisation of the density gives $f_x/f = r - s$. For a suitable expression F in the shape quantities, define

$$\mathcal{W}(F) = r\partial_x F + (v + r^2 - rs)F.$$

Then $f\mathcal{W}(F) = \partial_x(rfF)$. The smooth extensions avoid an interior boundary term at zero, and the source's differentiated Gaussian tail estimates give $rfF \to 0$ at both spatial infinities for the polynomial correction used below. Hence $\mathbb{E}_f \mathcal{W}(F) = 0$, where $\mathbb{E}_f$ denotes integration against $f(x)\,dx$.

The remaining step is algebra under these analytic restrictions. The correction polynomial below is chosen by first arranging a square and other nonnegative terms containing the highest derivative z, then cancelling their linear terms in z through the weighted divergence. The residual depends only on lower derivatives and the shape quantities. OpenAI (2026a), Lemma 4.1 supplies an explicit polynomial F_2 and a decomposition

$$z^2 - 12vw^2 + 6v^4 = \mathcal{W}(F_2) + \kappa v^4 + \mathcal{R}, \qquad \mathcal{R} \geq 0, \qquad \kappa = \frac{1661507}{1521000000} > \frac{1}{1000}.$$

The coefficients of F_2 have both signs; no sign of F_2 or $\mathcal{W}(F_2)$ is needed, since the correction integrates to zero. In contrast, the shape inequalities make every term of $\mathcal{R}$ nonnegative. Two of

its actual terms are

$$\frac{337}{10000}\, r^2 j\,(-w/r - 7v)^2, \qquad (2rH_x)v\left(\frac{2331}{10000}r^2 + \frac{87}{2500}j\right).$$

The first needs the joint comparison $j \geq 0$; the second also uses $H_x \geq 0$ on the positive half-line. Reflection preserves these signs on the negative half-line. The remaining terms use $H \leq 0$, forward concavity and the other shape bounds. Thus the decomposition is positive on the shapes supplied by the PDE, not on arbitrary spatial derivatives. Integration removes $\mathcal{W}(F_2)$ and keeps the fourth-moment margin, giving

$$\int_{\mathbb{R}} (z^2 - 12vw^2 + 6v^4) f\, dx \geq \frac{1}{1000}\int_{\mathbb{R}} v^4 f\, dx. \tag{5}$$

We use (5) as an imported analytic estimate. The full polynomial and its decomposition are in Lemma 4.1 and Table 1; Proposition 4.2 integrates them using the signs and zero boundary flux supplied by Lemma 3.4. Long §4.4, "How the signs make a correction with zero expectation useful" reproduces the definitions and complete identity.

To obtain Lemma 2 for an integrable coefficient, choose positive monotone step coefficients γ_n retaining the plateau near a fixed interior time, with $\|\gamma_n - \gamma\|_1 \to 0$, and terminal parameters $M_n \geq \|\gamma_n\|_\infty$ with $M_n \to \infty$. Spatial derivatives through order four converge uniformly on a fixed strip below one, and the diffusions driven by the same Brownian motion converge (see OpenAI (2026a), Lemmas 2.1–2.2 and Proposition 5.1). Only v, w, z remain in (5), so no convergence theorem for the auxiliary logarithmic derivative s is needed. Dividing by c^2 gives (3); strictness follows from $\Phi_{xx} > 0$ on the plateau (see OpenAI (2026a), Lemma 2.3). The plateau-preserving approximants and their common strip bounds are specified in long §4.5. The zero plateau has the separate heat identity in Lemma 2. Its strictness uses a positive diffusion density and the fact that Φ_{xxx} cannot vanish identically: an even quadratic profile would contradict the one-Lipschitz bound and $|x| + O(1)$ growth.

Passage to a fixed time. Now let $\gamma \in \mathcal{U}$ be continuous at $t \in (0,1)$, with $c = \gamma(t) > 0$. For $0 < \delta < \min(t, 1-t)$ define

$$\gamma_\delta(s) = \begin{cases} c, & t-\delta \leq s < t+\delta, \\ \gamma(s), & \text{otherwise.} \end{cases}$$

The inequalities $\gamma(t-\delta) \leq c \leq \gamma(t+\delta)$ make both joins nondecreasing. Thus $\gamma_\delta \in \mathcal{U}$; it need not minimise $\mathcal{P}$. Moreover,

$$\|\gamma_\delta - \gamma\|_1 \leq 2\delta\,[\gamma(t+\delta) - \gamma(t-\delta)] \longrightarrow 0.$$

Use a subscript δ for its value-function derivatives and diffusion. Lemma 2 and (4), already extended to integrable coefficients, give at the fixed time t

$$\mathcal{C}_{\gamma_\delta}(t) = \mathbb{E}[d_\delta^2 - 12c\, a_\delta b_\delta^2 + 6c^2 a_\delta^4] \geq \frac{c^2}{1000}\mathbb{E}[a_\delta^4]. \tag{6}$$

Choose $t < T < T_1 < 1$. Monotonicity gives the common coefficient bound

$$\sup_{s \leq T_1} \gamma_\delta(s) \leq \frac{\|\gamma_\delta\|_1}{1 - T_1}.$$

This is the first uniform estimate; the spatial derivative bounds require the smoothing theorem. Lemma 2.1 of OpenAI (2026a) gives uniform convergence in space and uniform bounds for $a_\delta, b_\delta, d_\delta$ on $[0,T]$, with the next spatial derivatives bounded as well. Lemma 2.2 couples the diffusions using the same Brownian motion and gives $\sup_{s\le T}|X^\delta_s - X_s| \to 0$. The derivative bounds control evaluation on these paths, so the integrands in (6) converge and are uniformly bounded. Dominated convergence passes their expectations to the limit. The common coefficient L^1 bound and the fixed distance from time one determine these estimates; the shrinking plateau width does not. The result is $\mathcal{C}_\gamma(t) \ge \gamma(t)^2\mathbb{E}a^4/1000$. This proves Proposition 3. No time derivative of the coefficient or minimising assumption has entered the limit.

A qualitative crossing criterion already suffices for a bounded gap: if $q''=0$ implies $q'''>0$, then q' cannot attain an interior maximum. The endpoint slope bounds therefore give $q'<1$ inside the gap and the same strict area deficit. Proposition 3 supplies more for the next argument: it applies at the chosen comparison time without assuming $q''=0$, and its fourth-moment lower bound survives the local replacement. Bare strict positivity could be lost in that limit.

4. A second use: the density cannot vanish inside the interval

Full support does not force pointwise positive density. For instance, $\gamma(t)=\int_0^t (s-\frac12)^2\,ds$ has full Stieltjes support but $\gamma'(\frac12)=0$: every interval of positive length has a positive integral, although the integrand vanishes at one point. The quantitative remainder in (5) rules this out for a minimiser. First, the identity $\mathbb{E}a^2=1$ expresses γ as a quotient of continuous moments and then gives smoothness. Only afterwards do we apply the fixed-time spatial estimate to obtain $\gamma'>0$.

Corollary 4 (smooth positive density; OpenAI (2026a), Corollary 6.4). Under the hypotheses of Theorem 1, $\gamma(0)=0$ and $\gamma\in C^\infty([0,1))$, with derivatives at zero taken from the right. Its Stieltjes measure is $\mu_\gamma(dt)=\gamma'(t)\,dt$, and

$$2\gamma'(t)\mathbb{E}[a(t,X_t)^3] \ge \frac{\gamma(t)^2}{1000}\mathbb{E}[a(t,X_t)^4] > 0 \qquad (0<t<1). \tag{7}$$

Only nonnegativity of the density at zero is asserted.

Proof. All spatial factors in expectations below are evaluated at (t,X_t). Since $da=-\gamma a^2\,dt+b\,dB$, Itô's formula applied to a^2 gives, for almost every t, $0=\frac{d}{dt}\mathbb{E}a^2=\mathbb{E}b^2-2\gamma\mathbb{E}a^3$. Here the left side vanishes because full support gives $\mathbb{E}a^2=1$. Solving for the coefficient yields

$$\gamma(t)=\frac{\mathbb{E}b^2}{2\mathbb{E}a^3}. \tag{8}$$

Initially (8) holds almost everywhere. Its denominator is at least two, because $a\ge 0$ and $\mathbb{E}a^3 \ge (\mathbb{E}a^2)^{3/2}=1$. The moments are continuous; right continuity of γ extends (8) to every time and proves continuity of γ. Evenness gives $b(0,0)=0$, hence $\gamma(0)=0$.

For smoothness, fix $T<1$ and consider all polynomial moments

$$M_P(t)=\mathbb{E}P(\Phi_x(t,X_t),\dots,\partial_x^m\Phi(t,X_t)),$$

where P is a polynomial in any finite number m of variables. Spatial differentiation of the PDE and Itô's product rule give polynomials Q_P, R_P, involving finitely many further spatial derivatives, such that

$$M_P(t)-M_P(0)=\int_0^t [M_{Q_P}(s)+\gamma(s)M_{R_P}(s)]\,ds.$$

No derivative of γ enters this identity. Bounded spatial derivatives on the fixed strip make every moment continuous. The integral identity then makes all moments C^1, and (8) makes γ C^1. Repetition proves smoothness. This is the quotient-and-moment argument of Chen (2026), Proposition 4.6 and Appendix B.6, using the polynomial-moment method of Auffinger and Chen (2015), Lemma 2 and the proof of Theorem 2(ii). The explicit polynomials and the passage from initially nonsmooth coefficients are in OpenAI (2026a), proof of Corollary 6.4 and long §9.1.

Now the two differentiated moments are

$$\frac{d}{dt}\mathbb{E}b^2 = \mathbb{E}[d^2 - 6\gamma ab^2], \qquad \frac{d}{dt}\mathbb{E}a^3 = \mathbb{E}[3ab^2 - 3\gamma a^4].$$

For a smooth time coefficient, differentiating $q'' = \mathbb{E}b^2 - 2\gamma\mathbb{E}a^3$ therefore gives

$$q^{(3)} + 2\gamma'\mathbb{E}a^3 = \mathcal{C}_\gamma(t). \tag{9}$$

This identity displays the two uses of the same spatial moment. On a plateau, $\gamma' = 0$ and $(v, w, z) = c(a, b, d)$ recover (4). For the now smooth minimiser, $q = t$, so $q^{(3)} = 0$ and the left side is $2\gamma'\mathbb{E}a^3$. Proposition 3 supplies the lower bound for this spatial moment.

Apply Proposition 3 to the now smooth minimiser. Strict increase and $\gamma(0) = 0$ give $\gamma(t) > 0$ for $t > 0$; (9) therefore gives (7). Strictness follows from $\gamma(t) > 0$ and $\mathbb{E}a^4 \geq (\mathbb{E}a^2)^2 = 1$. In particular, Cauchy–Schwarz gives

$$\mathbb{E}a^3 \leq \sqrt{\mathbb{E}a^2\,\mathbb{E}a^4} = \sqrt{\mathbb{E}a^4} \leq \mathbb{E}a^4,$$

so the same inequality implies $\gamma'(t) \geq \gamma(t)^2/2000$ for $0 < t < 1$.

Proposition 3 passes spatial moments, so its proof does not identify q_δ''' with the third derivative of the limiting moment. Indeed the minimiser has $q(t) = t$ and $q''' = 0$; the coefficient derivative in (9) is exactly what the constant-interval formula omits. Long §§9.1–9.2 retain the smoothness argument and fixed-time transfer, including the common approximation bounds.

Relation to earlier estimates. The backward shape comparisons, transformed density and positive-temperature gap-crossing argument appear in Lopatto (2026), v3, Propositions 3.6, 7.1 and 9.1. H.-B. Chen (2026), Proposition 4.2 proves the zero-temperature crossing implication $q'' = 0 \Rightarrow q''' > 0$ under the minimiser convention of §3.1. Lopatto's v4, Proposition 5.1 and Corollary 5.6 give qualitative crossing for arbitrary positive-temperature probability measures. In unit-noise coordinates, their cumulative coefficient is $\gamma_\beta = \beta\alpha_\nu$, where $\alpha_\nu(t) = \nu([0, t])$, and the terminal value is $\beta^{-1}\log\cosh(\beta x)$. Every bounded $\gamma \in \mathcal{U}$ has this form: choose $\beta > 0$ with $\beta \geq \sup\gamma$ and put the remaining probability mass at one. Thus the coefficient classes alone do not distinguish the estimates; the terminal datum and the quantitative control through its zero-temperature limit matter.

Lopatto's external-field paper, Theorem 2.3 and §8 also uses a fixed-time spatial inequality for support and positive density. At a continuity point, monotonicity supplies its one-sided coefficient bounds; on a smooth full-support interval of the minimiser, its scalar conditions become equalities. Appendix B gives the positive-temperature, zero-field strict-density result. The bound used here instead applies at a nonstationary comparison time and retains an explicit fourth-moment margin through the local replacement with terminal datum $|x|$. The rescaling and exact hypothesis comparisons are in long §2.3, finite-temperature dictionary.

5. Consequences for optimisation

5.1 A martingale formula for the ground-state value

The support identities give a scalar representation of P_*. Set $M_t = u(t, X_t)$, the bounded gradient martingale satisfying $dM_t = a(t, X_t)\, dB_t$. The quadratic variations distinguish the two Hessian moments:

$$d\langle M\rangle_t = a(t, X_t)^2\, dt, \qquad d\langle M, B\rangle_t = a(t, X_t)\, dt.$$

Full support gives $\mathbb{E}a(t, X_t)^2 = 1$, hence each increment $M_t - M_s$ has unconditional variance $t - s$. This normalises the discrete increments below. The first moment of a instead gives covariance with the driving noise. Let U be the almost-sure and L^2 limit of M_t. The value identity is

$$P_* = \mathbb{E}[UB_1] = \int_0^1 \mathbb{E}[a(t, X_t)]\, dt. \tag{10}$$

Boundedness gives $|U| \le 1$, while $\mathbb{E}M_t^2 \to 1$ gives $\mathbb{E}U^2 = 1$; therefore $U^2 = 1$ almost surely. To see the cancellation behind (10), the terminal convergence of the convex value functions gives $UX_1 = |X_1|$ (the equality is automatic on $X_1 = 0$), while Itô's formula gives $\mathcal{P}(\gamma) = \mathbb{E}|X_1| - \int_0^1 t\gamma(t)\, dt$. Multiplying $X_1 = B_1 + \int_0^1 \gamma(t)M_t\, dt$ by U cancels that integral because $\mathbb{E}[UM_t] = t$. The Parisi formula then gives (10). The endpoint limits and stochastic-integral covariance are proved in OpenAI (2026a), Corollary 7.1 and the long record, §6, "Recovering the value from a martingale". This argument uses full support, not the density corollary. In particular, convexity and Cauchy–Schwarz give $0 \le \mathbb{E}a \le 1$, so

$$0 \le P_* - \int_0^T \mathbb{E}[a(t, X_t)]\, dt \le 1 - T.$$

One can therefore fix the terminal cutoff before discretising the diffusion.

The gradient-martingale and value methods have antecedents in Montanari, preprint v2, Lemmas 3.2–3.4, under its positive-temperature support assumption, and in the zero-temperature calculations of El Alaoui, Montanari and Sellke, preprint v1, Lemmas 6.5 and 6.13. Sellke, preprint v4, Proposition 3.10 and Lemma 3.11, p. 17 gives the closely related finite-cutoff cancellation by applying Itô's formula to Φ and $x\Phi_x$.

5.2 The classical algorithm

Full support and existence give an attained, strictly increasing minimiser in the monotone zero-temperature class. This is Assumption 2 of El Alaoui, Montanari and Sellke, preprint v1. Their Corollary 2.2 then gives, for every fixed $\varepsilon > 0$, a randomised algorithm returning a configuration within εn of the finite SK maximum with probability tending to one over couplings and algorithmic randomness. Its cost is $C(\varepsilon)n^2$ real-arithmetic operations with the prescribed scalar functions and normalisers available; this includes no separate numerical-preprocessing or finite-precision bound. OpenAI (2026a), Corollary 7.3 and long §9.4 check the Gaussian convention and operation count. Chen already records the classical consequence in §1, p. 4. In their incremental approximate message-passing construction, the noise and drift coefficients are, in our notation,

$$u_{\mathrm{EAMS}}(t, x) = a(t, x), \qquad v_{\mathrm{EAMS}}(t, x) = \gamma(t)u(t, x).$$

Thus their noise multiplier is the Hessian, whereas our u is the gradient. The bounded martingale M_t and $\mathbb{E}a^2 = 1$ give the required variance normalisation. With the fixed-strip regularity and

discretisation estimates, their Theorem 2 achieves energy arbitrarily close to the truncated first-moment integral in (10). Strict positivity of the density is unnecessary.

5.3 A fixed QAOA circuit before the thermodynamic limit

The remaining consequence uses a separate circuit approximation theorem, stated as Proposition 5 below. We give its finite Gaussian input and identify the energy it approximates. Its operator and control construction is developed in the companion; the support and density proofs above do not require those additional prerequisites.

Let $\widehat{Z}_i, \widehat{X}_i$ be Pauli operators, and define the cost and mixer by

$$C_{n,J} = \frac{1}{\sqrt{n}} \sum_{i<j} J_{ij} \widehat{Z}_i \widehat{Z}_j, \qquad B_n = \sum_i \widehat{X}_i.$$

The cost eigenvalues are $H_n(\sigma)$. QAOA alternates cost evolution and mixing from the uniform state $|+\rangle^{\otimes n}$ (Farhi, Goldstone and Gutmann (2014)). For a fixed sequence $W = (\theta_1, \beta_1, \dots, \theta_p, \beta_p)$, write

$$|\psi_{n,J}(W)\rangle = e^{-i\beta_p B_n} e^{-i\theta_p C_{n,J}} \cdots e^{-i\beta_1 B_n} e^{-i\theta_1 C_{n,J}} |+\rangle^{\otimes n},$$

$$v(W) = \lim_{n\to\infty} \frac{1}{n} \mathbb{E}_J \langle \psi_{n,J}(W) | C_{n,J} | \psi_{n,J}(W) \rangle.$$

Farhi, Goldstone, Gutmann and Zhou (2022) established this fixed-angle limit. OpenAI (2026b), Theorem 1.1 proves that for every $\varepsilon > 0$ there is one finite W with $v(W) \geq P_* - \varepsilon$, whose depth and angles depend on ε but not on n or J. The instance still determines $C_{n,J}$. Here depth counts global cost–mixer layers; their elementary gate counts depend on the implementation. This order of limits differs from the fixed-instance depth limit in Farhi, Goldstone and Gutmann, §VI, which permits instance-dependent choices. The common-circuit conclusion was conjectured by Basso, Farhi, Marwaha, Villalonga and Zhou (2022), §7, equation (7.2).

The finite Gaussian input. Fix the minimiser and its diffusion, a cutoff $0 < T < 1$, and a mesh $t_j = j\Delta$, $\Delta = T/N$. The mesh count N is distinct from the number of spins n. From independent standard normals $Z_1, \dots, Z_{N+1}$, form

$$\begin{aligned} Y_0 &= 0, \\ Y_j &= Y_{j-1} + \sqrt{\Delta} Z_j + \Delta \gamma(t_{j-1}) u(t_{j-1}, Y_{j-1}), \\ h_j &= \frac{a(t_{j-1}, Y_{j-1})}{\sqrt{\mathbb{E} a(t_{j-1}, Y_{j-1})^2}}, \qquad g_j = h_j Z_j \quad (1 \leq j \leq N). \end{aligned}$$

OpenAI (2026a), Proposition 7.2 proves that the denominators are positive for all sufficiently large N. Since h_j depends only on $Z_1, \dots, Z_{j-1}$, the g_j are centred and orthonormal in L^2. They need not be independent Gaussians. With Φ_{G} the standard normal distribution function, set

$$\begin{aligned} F_N &= \operatorname{sgn}\{u(T, Y_N) + 2\Phi_{\mathrm{G}}(Z_{N+1}) - 1\}, \\ \mathbb{E}[F_N \mid Z_1, \dots, Z_N] &= u(T, Y_N). \end{aligned}$$

The second equality follows because $\Phi_{\mathrm{G}}(Z_{N+1})$ is uniform on $[0, 1]$: this random rounding preserves every correlation with a function of the preceding coordinates. The finite quantity needed by the

circuit construction is the adjacent pairing

$$S_{T,N} := \sum_{j=1}^{N} \mathbb{E}[F_N g_j]\,\mathbb{E}[F_N Z_{j+1}],$$
$$\lim_{N\to\infty} S_{T,N} = I_T := \int_0^T \mathbb{E}[a(t, X_t)]\,dt. \tag{11}$$

The same proposition proves this limit. The Euler approximation and the gradient martingale give

$$\left\| u(T, Y_N) - \sum_{j=1}^{N} \sqrt{\Delta}\, a(t_{j-1}, Y_{j-1}) Z_j \right\|_2 \longrightarrow 0.$$

Conditional-mean preservation transfers these correlations to F_N for the first N coordinates. Rounding itself has squared L^2 error $\mathbb{E}(F_N - u(T, Y_N))^2 \to 1-T$, so it is not the vanishing residual. Bessel's inequality, applied separately to (g_j) and (Z_j), controls the errors of both correlation vectors in ℓ^2. Their bounded norms control the growing pairing by Cauchy–Schwarz. The terms with $j < N$ give a Riemann sum, and the last rounding-coordinate term vanishes. All these limits keep T fixed; only afterwards does $T \uparrow 1$ give P_*. The construction and estimates are in long §7, "A finite Gaussian spin and its adjacent correlations".

Proposition 5 (circuit approximation; a consequence of OpenAI (2026b), §§2–5). For each fixed T, N in the preceding construction with finite positive normalisers, and every $\eta > 0$, there is a finite ordinary QAOA word W, with deterministic angles independent of n and J, such that

$$|v(W) - S_{T,N}| < \eta.$$

Why the adjacent pairing is the energy. We use the rooted Gaussian representation of OpenAI (2026b), §2 and Lemma 3.1, building on the spin–boson description of Boulebnane et al. (2026). Its classical sector consists of square-integrable functions built recursively from a local Gaussian seed and child-generated Gaussian coordinates, with inner product given by expectation. A centred real child direction f gives a Gaussian coordinate $\mathsf{G}(f)$ at the receiving root, with

$$\mathbb{E}[\mathsf{G}(f)\mathsf{G}(g)] = \langle f, g\rangle.$$

The receiving root's seed is independent of these coordinates; compatible embeddings preserve them as the recursive depth grows. The root shift L inserts a child direction into the first Gaussian chaos (the linear Gaussian part) with constant receiving-root state. Its adjoint projects onto that part and returns the child direction. The depth-one calculation in long §8.1.1 checks this representation's energy against the finite SK circuit before the general construction in §8.1.

Start with a standard normal receiving-root seed g_0 and recursively put $Z_{j+1} = \mathsf{G}(g_j)$. The orthonormal child directions give independent standard Gaussian lifts, reproducing the scalar law. The finite spin F_N depends only on those lifts. Projection therefore gives the exact rooted-energy identity

$$\mathsf{L}^* F_N = \sum_{j=0}^{N} \mathbb{E}[F_N Z_{j+1}] g_j,$$
$$\mathcal{E}(F_N) := \mathrm{Re}\langle F_N, \mathsf{L} F_N\rangle = \sum_{j=0}^{N} \mathbb{E}[F_N g_j]\,\mathbb{E}[F_N Z_{j+1}] = S_{T,N}. \tag{12}$$

Independence of the receiving-root seed removes the $j = 0$ term. This identity concerns the finite Gaussian formula in a limiting rooted space, not a finite-n circuit. The unshifted pairing can have the same scalar limit without identifying this energy. The construction and projection are proved in long §8.1.

Construction underlying Proposition 5. Smooth approximation of the final sign, followed by Lemma 3.2, gives a bounded smooth odd formula with bounded transmitted functions; smoothing the sign alone is insufficient. Definition 3.3 and Lemma 3.4 then partition the seed into classes and exclude the receiving seed class, its path-ancestor classes and all reserved helper classes from each dependency, preventing return through an already traversed branch. The output is zero on helper classes, one per positive aggregation depth, whose total probability is small but positive during synthesis.

Theorem 4.1 realises the prepared formula with temporary Gaussian controls, reusing the same sitewise seeds throughout the word. Theorem 5.1 removes these controls with arbitrarily small energy error, and Proposition 2.1 identifies the fixed ordinary word's limiting energy with $v(W)$. All these errors are absolute, so the argument applies to each fixed $S_{T,N}$, before it is chosen close to P_*. The output preserved by Proposition 5 is energy; the measured-configuration conclusion below uses a further concentration theorem. The complete error bound is in long §8.6, finite-formula energy approximation, with the constructions and their hypotheses in long §§8.1–8.6.

The deduction of near-optimality is now explicit:

$$P_* - v(W) \le (P_* - I_T) + |I_T - S_{T,N}| + |S_{T,N} - v(W)|. \tag{13}$$

Choose T so that $1 - T < \varepsilon/3$, then N so that the second term is below $\varepsilon/3$, and finally use Proposition 5 with $\eta < \varepsilon/3$. These finite choices give $v(W) > P_* - \varepsilon$ before n tends to infinity. The source's Appendix A supplies an alternative positive-temperature input to the same synthesis. The selected construction supplies no quantitative depth bound, finite-size rate or efficient procedure for finding angles.

Corollary 6 (a measured configuration on the same instance). For every $\varepsilon > 0$ there is a finite ordinary QAOA word W_ε, chosen independently of n and J, such that one computational-basis measurement Σ satisfies

$$\Pr_{J,\Sigma}\Big\{\max_{\sigma} H_{n,J}(\sigma) - H_{n,J}(\Sigma) \le \varepsilon n\Big\} \longrightarrow 1 \qquad (n \to \infty). \tag{14}$$

Proof. For a fixed word, put $Y_n = H_{n,J}(\Sigma)/n$ and $K_n = \max_\sigma H_{n,J}(\sigma)/n$. Farhi, Goldstone, Gutmann and Zhou, §4, equations (13)–(22) prove

$$\mathbb{E}_{J,\Sigma} Y_n \to v(W), \qquad \mathbb{E}_{J,\Sigma} Y_n^2 \to v(W)^2.$$

The second moment is $\mathbb{E}_J\langle (C_{n,J}/n)^2\rangle$, so it includes measurement variance as well as disorder fluctuations. Thus $Y_n \to v(W)$ in joint L^2. Gaussian concentration of the maximum and its defining mean limit give $K_n \to P_*$ in probability. Choose W with $P_* - v(W) < \varepsilon/3$. With probability tending to one, both $K_n - P_* < \varepsilon/3$ and $v(W) - Y_n < \varepsilon/3$, which proves (14). The two energies use the same J; no independence between them is needed. The companion's §11.2, "What is averaged, and a fixed-word probability consequence" gives the concentration argument and its further probability consequences.

Formal scope

This AI-assisted article reconstructs the cited analytic arguments. The integrated estimate (5), its shape hypotheses and compact-strip approximation are substantial imported inputs.

The full-support formal statement asserts the support and moment conclusions conditional on an admissible minimiser, and constructs its associated diffusion. A separate scalar formal statement covers value identities and finite Gaussian approximation, with minimiser and diffusion premises for those clauses. Its third clause identifies the ground-state and Parisi values from a minimiser, constructing the diffusion internally. The two formal models have not been transported into one another in this record. Neither statement includes the positive-density corollary or QAOA convergence. Their full-support and scalar implementations use Lean and mathlib.

Acceptance by Comparator and extraction of the selected formal statements remain unknown. Source inspection alone establishes neither. Exact versions, checking outcomes and correspondence are retained in the long record's §12, "What the formal statements establish" and §13, "Source checks and finite calculation".

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